

Vehicle Classification Project

Use Case 1: Vehicle Classification for Parking Management

Objective

To develop an AI-powered vehicle classification system that optimizes parking space allocation, automates fee calculations, and improves security in smart parking lots.

Methodology

1. **Data Collection** – Capture images of vehicles entering/exiting parking lots.
2. **Preprocessing** – Adjust brightness, contrast, and remove noise.
3. **Model Selection** – Train a CNN-based model for vehicle classification.
4. **Feature Extraction** – Identify vehicle type, size, and license plate.
5. **Classification** – Categorize vehicles into small, medium, or large types for parking allocation.
6. **Integration** – Link the classification results to parking fee calculation and security alerts.

Dataset

- **Source:** Real-world parking lot footage and open datasets (PKLot, CNRPark).
- **Attributes:** Vehicle type, license plate, parking duration, entry/exit timestamps.
- **Size:** Thousands of images covering various vehicle models and angles.

Expected Outcomes

- Smart allocation of parking spots based on vehicle size.
- Automated parking fee calculations based on vehicle type and duration.
- Improved security through vehicle tracking and anomaly detection.
- Reduction in manual intervention and operational costs.

Dataset : <https://www.kaggle.com/datasets/ammarnassanalhajali/pklot-dataset>

Use Case 2: Autonomous Vehicle Perception and Recognition System

Objective

To enable autonomous vehicles to recognize and classify surrounding vehicles, enhancing decision-making for safe navigation.

Methodology

1. **Data Collection** – Collect vehicle images from onboard cameras and LiDAR sensors.
2. **Preprocessing** – Convert raw sensor data into structured inputs.
3. **Model Selection** – Use deep learning models (YOLO, Faster R-CNN) for classification.
4. **Feature Extraction** – Detect vehicle size, shape, speed, and movement patterns.
5. **Classification** – Categorize vehicles into different types for real-time decision-making.
6. **Integration** – Feed classification results into autonomous driving systems for adaptive behaviour.

Dataset

- **Source:** Open datasets (KITTI, nuScenes, Waymo) and real-world sensor data.
- **Attributes:** Vehicle type, distance, speed, direction, position relative to ego-vehicle.
- **Size:** Large dataset including urban, highway, and off-road scenarios.

Expected Outcomes

- Enhanced perception capabilities for self-driving cars.
- Improved decision-making in traffic situations.
- Reduced collision risks by predicting vehicle behaviour.
- Better interaction with human-driven vehicles for smoother navigation.

Dataset : <https://www.kaggle.com/datasets/klemenko/kitti-dataset>

Use Case 3: AI-Powered Roadside Assistance Dispatch

Objective

To develop an AI-driven system that detects and classifies vehicles in distress, enabling roadside assistance providers to dispatch the appropriate help efficiently and reduce response times.

Methodology

1. **Data Collection** – Capture images and videos of stranded vehicles via roadside cameras, drones, and user-submitted mobile photos.
2. **Preprocessing** – Enhance image clarity by adjusting brightness, contrast, and removing obstructions.
3. **Model Selection** – Train a deep learning model (e.g., ResNet, YOLO) for vehicle identification.
4. **Feature Extraction** – Identify vehicle type, make, model, and possible damage extent.
5. **Classification** – Categorize vehicles into standard classifications (sedan, SUV, truck, motorcycle, etc.).
6. **Integration** – Transmit classification results to dispatch systems for optimal service assignment (tow truck, repair vehicle, fuel delivery).

Dataset

- **Source:** Real-world roadside assistance reports, open datasets (Stanford Cars, Vehicle Damage Dataset).
- **Attributes:** Vehicle type, condition (damaged/undamaged), location, assistance request type.
- **Size:** Thousands of labelled vehicle images across different roadside breakdown scenarios.

Expected Outcomes

- Faster response times by assigning appropriate assistance based on vehicle type and condition.
- Enhanced accuracy in dispatching services like towing, tire replacement, or fuel delivery.
- Improved customer satisfaction through automated vehicle recognition for quicker service.
- Reduction in operational costs by optimizing dispatch logistics.

Dataset : <https://www.kaggle.com/datasets/lplenka/coco-car-damage-detection-dataset>

Use Case 4: Intelligent Emissions Monitoring System

Objective

To leverage AI for identifying and classifying vehicles based on fuel type and emissions output, aiding in environmental impact analysis and regulatory enforcement.

Methodology

1. **Data Collection** – Gather vehicle images from traffic cameras, toll booths, and emissions testing centres.
2. **Preprocessing** – Improve image clarity, remove obstructions, and standardize inputs.
3. **Model Selection** – Train a deep learning model to classify vehicles based on fuel type (gasoline, diesel, electric, hybrid).
4. **Feature Extraction** – Identify vehicle type, manufacturer, fuel type, and emissions level.
5. **Classification** – Categorize vehicles based on emissions data and flag high-pollution vehicles.
6. **Integration** – Provide classification results to city authorities for emissions control and eco-friendly policymaking.

Dataset

- **Source:** Public emissions datasets, vehicle registration records, real-world traffic camera feeds.
- **Attributes:** Vehicle type, fuel type, emission level, registration data.
- **Size:** Large dataset covering urban and highway environments with diverse vehicle models.

Expected Outcomes

- Real-time identification of high-emission vehicles for regulatory enforcement.
- Promotion of electric and hybrid vehicle adoption through emissions monitoring.
- Improved urban pollution tracking and air quality analysis.
- Data-driven policymaking for eco-friendly transport regulations.

Dataset : <https://www.kaggle.com/datasets/s3programmer/vehcle-emission-dataset>

Use Case 5: Smart Fleet Management for Delivery Vehicles

Objective

To create an AI-powered system that classifies and monitors delivery vehicles in real-time, optimizing route planning, maintenance scheduling, and fuel efficiency for fleet operators.

Methodology

1. **Data Collection** – Capture images and sensor data from vehicles in the fleet (GPS, cameras, fuel consumption data).
2. **Preprocessing** – Clean, normalize, and filter data to ensure quality and consistency.
3. **Model Selection** – Use deep learning models (e.g., YOLO, ResNet) for vehicle classification based on type, condition, and fuel efficiency.
4. **Feature Extraction** – Identify vehicle make, model, fuel consumption, and maintenance history.
5. **Classification** – Categorize vehicles based on type (e.g., light truck, heavy truck, electric vehicle) and assess maintenance needs.
6. **Integration** – Feed classified data into fleet management systems for route optimization, predictive maintenance, and fuel management.

Dataset

- **Source:** Real-time fleet data, open datasets on vehicle performance (e.g., Fleet Maintenance Dataset, Vehicle Fuel Consumption Dataset).
- **Attributes:** Vehicle type, fuel consumption, GPS location, route history, maintenance logs.
- **Size:** Thousands of data points from a large fleet of diverse vehicles across various routes and terrains.

Expected Outcomes

- Optimized delivery routes for better fuel efficiency and reduced emissions.
- Proactive maintenance scheduling based on vehicle condition data.
- Real-time monitoring of fleet performance and fuel consumption.
- Reduced operational costs through data-driven decision-making in fleet management.

Dataset : <https://www.kaggle.com/datasets/imtkaggleteam/fleet-vehicles-fuel-consumption>